From phone photo to packshot: capture, edit, select

Team AACtivEtion Function — TEAM_MEMBER_1, TEAM_MEMBER_2, TEAM_MEMBER_3, TEAM_MEMBER_4, TEAM_MEMBER_5 Sartiq challenge, Neural Wave 2026, USI Lugano, 16–18 October 2026

1. Problem and approach

Sartiq wants a studio flat-lay packshot of a garment from an amateur smartphone photo. The official score compares our picture with Sartiq's studio reference: 40 % colour (global Lab distribution and a 4×4 spatial colour map), 25 % perceptual similarity (LPIPS), 15 % semantic similarity (CLIP), 15 % silhouette overlap and 5 % structural similarity. Two things follow: colour and silhouette dominate, and the picture must be *this* garment, not a plausible garment.

We did not train a model. Our contribution is the loop around a hosted image-editing model (Google Nano Banana 2.1, `google/nano-banana-2.1/edit` on fal.ai):

1. **Capture protocol.** Garments are laid flat on a light surface, smoothed and made symmetric, and shot top-down with the phone, plus 3–6 close-ups of collar, logo, print, buttons and fabric (`docs/CAPTURE_PROTOCOL.md`).
2. **Multi-image edit.** One request per candidate carries the front photo *and* the close-ups with a fixed prompt: reproduce this exact garment as a flat-lay on white, keep colour, texture, print, logo and proportions. The prompt tells the model what the extra images are for. Per-garment notes (`garment_notes.json`) name the details that must survive ("round black-on-cream badge reading 'Pretty Green'", "red Levi's tab on the pocket edge").
3. **Candidates and selection.** We generate three candidates per garment and pick one with a simple rule: the garment's median Lab colour must agree with the other candidates (colour consensus, because the model's colour errors are random from call to call), and the silhouette must be symmetric and upright (left/right mask symmetry and tilt of the principal axis). A human can override the pick (`manual_picks.csv`).
4. **Always submit.** For garments we could not photograph ourselves, a picture generated from the organisers' snapshot fills the slot.

Everything runs on a laptop in minutes; a full test-set run costs about USD 3 in API calls.

2. How we measured

The organisers' leaderboard gives one number per round, so we built an honest local loop first: a 38-pair holdout carved from `train/` with the official scorer (`scripts/make_holdout.py`, `--group-by-target`). Every idea was scored there before it touched the dev set. For the 15 on-site garments, which have no public reference, we scored each result against our own clean protocol photo as a stand-in reference ("fidelity check"); this is not the official score, but it ranks variants of the same garment consistently and flags shape errors.

3. What worked and what did not

Variant (38-pair holdout, official scorer)	Score
Hosted baseline (organisers' prompt, 1 candidate)	60.55
Our packshot prompt v2	61.01

Variant (38-pair holdout, official scorer)	Score
Prompt v2 + 3 candidates + colour-consensus selection	**61.04**
Constant Lab colour calibration fitted on 40 train pairs	60.70
Colour correction towards the phone photo's garment colour	59.06
Post-processing (flat-field, white background clean-up, unsharp)	≤ raw output

Findings that shaped the final pipeline:

- **Colour errors are random, not biased.** Across garments the model's median L error is ≈ 0 with a mean absolute error of ~5; a constant calibration therefore gains little, and correcting towards the phone photo *hurts* because the phone's white balance is worse than the model's. Spatial colour placement is already good (0.93); the loss is in the global colour, i.e. in the garment's hue being slightly off. Picking the candidate that agrees with the others on colour is the cheapest way to reduce that variance.
- **Shape follows the input lay.** Amateur flat-lay inputs score lowest when the lay is bad, because the model copies it; they score highest when the lay is straight. Prompt instructions about geometry ("make it narrower", "straighten") *reduced* the shape score (the model redesigned the garment); instructions about details ("buttoned flap pocket", "red tab") improved fidelity. We kept the detail notes and dropped the geometry ones.
- **Close-ups matter.** With the close-ups in the request, logos and prints are reproduced instead of paraphrased; the fidelity check on the on-site garments moved from ~50 (organisers' snapshots) to ~72 (our protocol shots + close-ups), with no `review`/`invalid` flags.

On the dev leaderboard the pipeline went from 68.31 (organisers' photos, one candidate, first prompt) to **70.79** (final prompt, three candidates, consensus selection), rank 1 at the last round we saw. Two late experiments — per-garment detail notes for the dev set and the model's "high" thinking level — did not beat 70.79 within noise, so the hand-in uses the 70.79 pictures.

4. Limitations and next steps

- The selection rule has no reference; it rewards agreement, not correctness. Three candidates is the minimum for a consensus; more candidates improve the odds at linear cost.
- Fine lettering is still the weakest point (small text on labels is sometimes paraphrased).
- The 10–13 point dev→test gap the organisers observed in rehearsal applies to us too; our on-site protocol is designed to close part of it.
- Next: a vision-model check that compares each candidate with the close-ups detail by detail (logo text, button count) and rejects candidates that drop a detail; a second edit pass restricted to the region of a wrong logo.

5. Reproducibility

Repository `NW_26_aactivetion_function`: `README.md` (run instructions), `pipeline.py`, `notebooks/day2_final_pipeline_v2.ipynb` (the exact cells we ran), `docs/CAPTURE_PROTOCOL.md`, `submissions/aactivetion_function/` (hand-in), `ui/` (jury demo). Dataset and API key are not in the repository.